Daniel Cid Brady

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Education

Georgia Institute of Technology **Expected May 2028**
Bachelor of Science in Electrical Engineering — GPA: 4.00/4.00 *Atlanta, GA*

Experience

AVL Technologies **Summer 2026**
Software and Controls Intern *Asheville, NC*

- Reworked PIC32 firmware in C/C++ using device manuals and MPLAB X/XC32; established communication between two controllers supporting a three-axis satellite antenna system.
- Traced Ethernet, motor-control, sensor, and RF receiver interfaces; developed an ADC test project and isolated firmware, device-programming, linker, and board-level faults.

Duke Energy **Spring 2026**
Electrical Engineering Co-op — Transmission Substation Engineering *Raleigh, NC*

- Revised one-line, three-line, planning, plan/section, and conduit/cable drawings in MicroStation for a 230 kV breaker replacement; incorporated multidisciplinary engineering feedback.
- Applied Duke Energy design standards and equipment requirements to prepare a transmission-substation design package for formal engineering review.

ABB Case Study Competition **April 2025**
Sales Engineer — First Place (1 of 7 Teams)

- Conducted market research and ROI analyses, evaluated technical product solutions, and synthesized findings into a four-week proposal and presentation for ABB professionals.

Projects

Current-Starved CMOS Ring Oscillator **Fall 2026**
Analog IC Design and Physical Layout *ECE 1100 Discovery Project*

- Designed and simulated a five-stage current-starved CMOS ring oscillator in Cadence Virtuoso targeting 500 MHz at 1.8 V with a buffered 1 pF output; completed physical layout and passed DRC and LVS checks.

MNIST Handwritten-Digit Classifier **Fall 2026**
C++ Neural Network Project *Georgia Tech*

- Implemented a two-layer C++ neural network with image/label loading, forward propagation, tanh activation, loss calculation, and gradient-descent updates using provided backpropagation routines.

Wireless Channel Modeling **2026**
C++ Programming Project *Georgia Tech*

- Built a C++ complex-number class and wireless-channel model to calculate signal/noise power, SNR, and modulation-dependent communication performance.

Additional Experience & Involvement

The Inn on Biltmore Estate — Engineer I Summer 2025

- Performed preventive and corrective maintenance across hotel systems and resolved guest-room work orders while meeting safety, reliability, and service standards.

Independent Tutoring — Calculus & Physics Tutor 2025 – Present

- Translate abstract technical concepts into clear explanations and adapt problem-solving approaches to individual learning styles.

Asheville Racquet Club — Tennis Camp Manager Summer 2023

- Managed daily camp operations, coordinated activities, and communicated with participants and staff.

Memberships: Georgia Tech Swim Club; Society of Sales Engineering.

Skills

Programming: C, C++, Python, R (advanced), MATLAB; Git, Linux/UNIX
Embedded / Hardware: PIC32 firmware, ADC testing, hardware/software troubleshooting, RF systems
Engineering Tools: MPLAB X, XC32, Cadence Virtuoso, MicroStation, AutoCAD, SolidWorks, DipTrace
Circuit Design / Analysis: CMOS circuits, transistor-level simulation, physical layout, DRC/LVS, technical documentation